

TrES-5b

R-1661

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_200813 · created 2026-08-02T09:31:25+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459074.752 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.053
Transit depth (Rp/Rs)^2	2.92 +/- 1.8 [%]
Orbital Inclination (inc)	89.21 +/- 2.94
Airmass coefficient 1 (a1)	10.6 +/- 6.02
Airmass coefficient 2 (a2)	-1.94 +/- 1.53
Scatter in the residuals of the lightcurve fit is	63.48 %
Best Comparison Star	None
Optimal Aperture	1.34
Optimal Annulus	11.59
Transit Duration (day)	0.0862 +/- 0.0069

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.171 ± 0.053 vs 0.142 (deviation 0.55σ)
Duration fit vs published	0.0862 d vs 0.0758 d (deviation 1.5σ)
Mid-time O–C	-7.8 min
Depth SNR	3.2
Residual scatter	63.48 %

Light curve

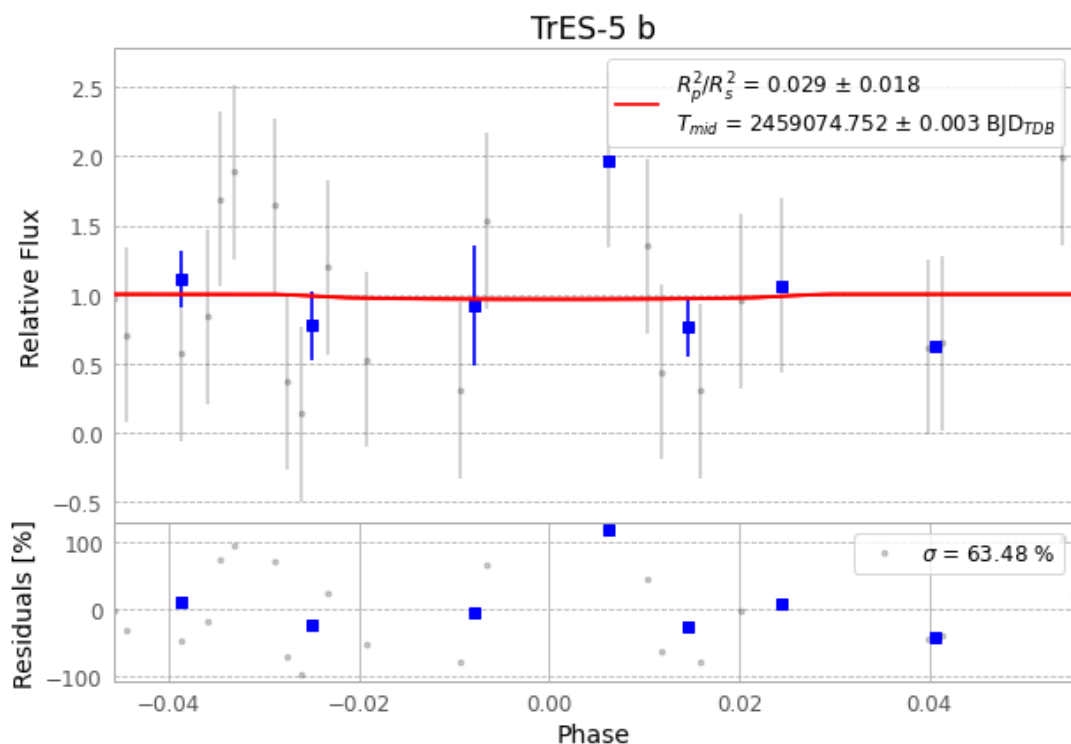


Figure 1 — FinalLightCurve_TrES-5 b_2020-08-12.png (SHA-256 ff71f9aa1354ab4b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [601, 228], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9160a605afa1a4aa...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), TRES-5200813042112.FITS → TRES-5200813080912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-200813.log (81a52ce76fa7...), FinalLightCurve_TrES-5 b_2020-08-12.png (ff71f9aa1354...), FinalLightCurve_TrES-5 b_2020-08-12.csv (8d18dde1cf19...)

Compute resources

Wall time 250.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 09:31Z.